

a camera image localized in a semantic 3DGS map

USGLoc → pose + covariance Σ
a filter can weight each frame by its own Σ

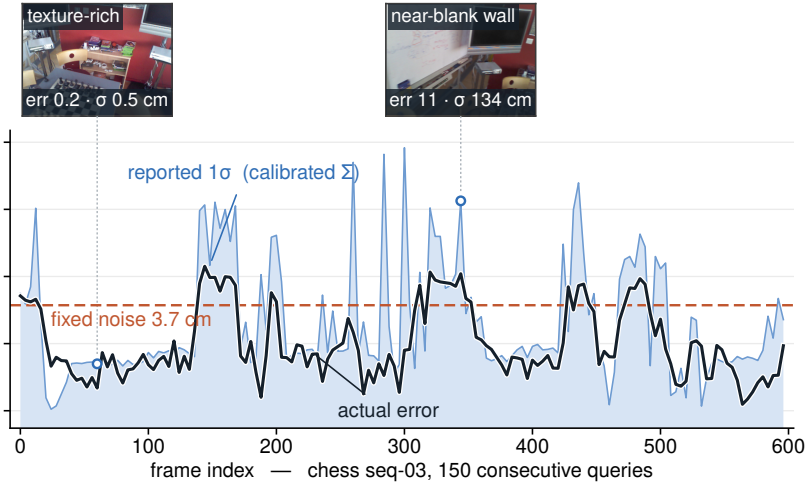
baselines → pose only
the filter must assume one hand-tuned noise

error spread by each method's own confidence
least- vs. most-confident decile, 500 queries

USGLoc σ — 0.26 → 11.5 cm — **44×**

ACE inliers — **7×** — 0.49 → 3.2 cm

accuracy is on par — 7-scene median 0.90 cm vs. ACE 1.05 cm; the calibrated covariance is what is new.



σ spans 0.18–109 cm (p5–p95, ×597); 71% of frames lie inside 1σ, 90% inside 2σ.
Without Σ, one fixed noise is ≥3× too large on 60% of frames, ≥3× too small on 21%.